INCH POUND

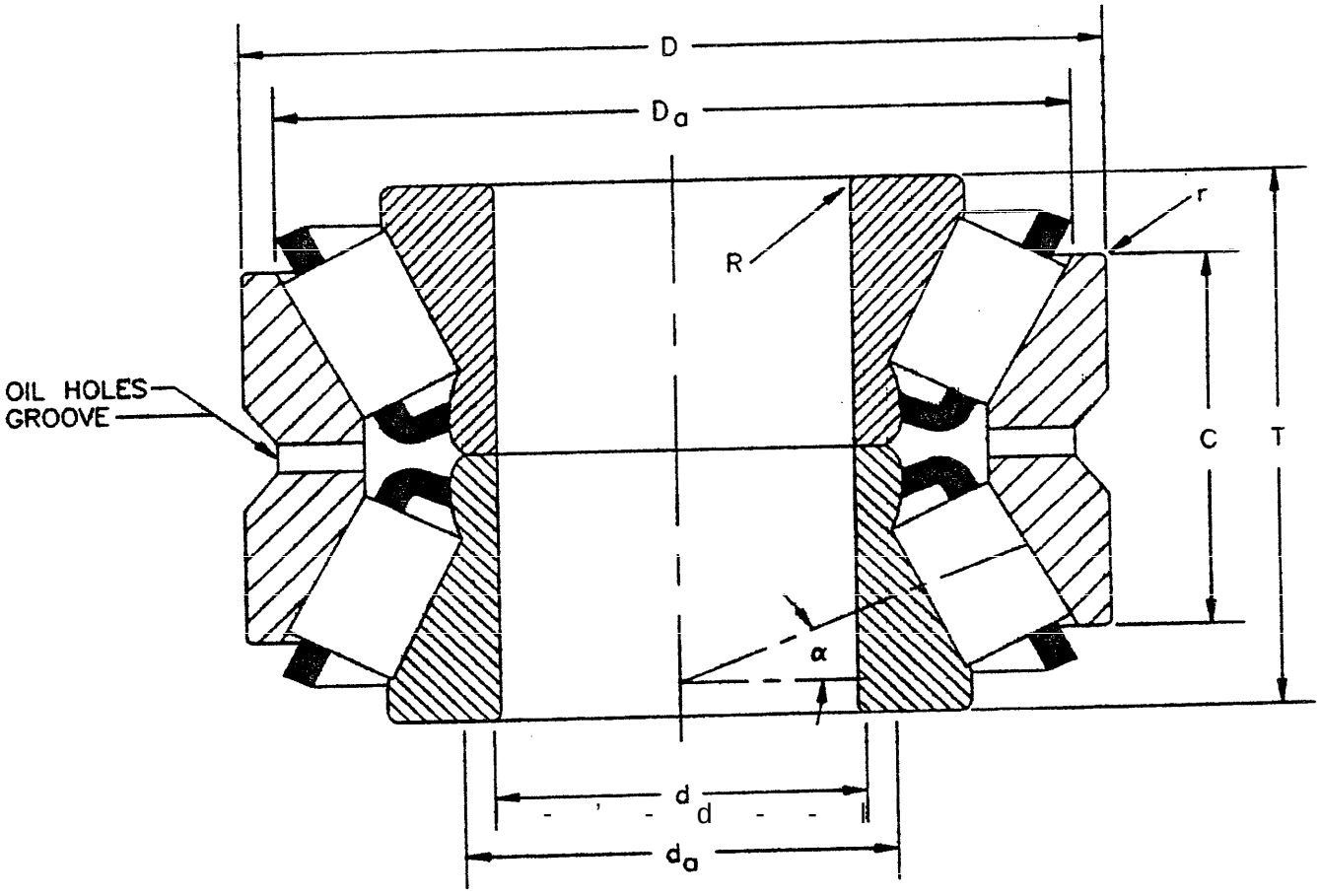
FF-B-187/8
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, DOUBLE ROW NORMAL ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP, NONADJUSTABLE. (TYPE TNA)

This specification is approved by the Administrator, General Services Administration, Federal Supply Service, for the use of all Federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist of this document and the latest issue of FF-B-157/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
01 *	NA05075-05185D	0.7500	1.8504	0.9926	1.2500	0.05	0.03	1.00	1.67	1.64	7500
02 *	NA08125-08231D	1.2500	2.3125	0.9688	1.2812	0.06	0.02	1.52	2.17	1.23	8900
03 *	NA13687-13621D	1.5000	2.7170	1.5000	1.8124	0.08	0.03	1.83	2.56	1.45	15400
04 *	NA438-432D	1.7500	3.7500	2.0000	2.4376	0.14	0.03	2.24	3.43	2.05	35200
05	NA366-363D	1.9685	3.5433	1.6563	1.9789	0.14	0.03	2.40	3.31	1.83	14000
06 *	NA3780-3729D	2.0000	3.6718	2.0625	2.5625	0.14	0.03	2.52	3.46	1.73	19700
07	NA455-452D	2.0000	4.2500	2.1250	2.5626	0.14	0.03	2.56	3.94	1.74	21800
08	NA539-533D	2.1250	4.3750	2.5000	3.1250	0.14	0.06	2.68	3.94	1.97	27600
09	NA385-384D	2.1654	3.9370	1.6875	2.0625	0.14	0.03	2.64	3.66	1.65	14800
10 *	NA558-552D	2.3750	4.8750	2.5000	3.1250	0.14	0.06	2.99	4.53	1.69	30700
11	NA569-563D	2.6250	5.0000	2.5625	3.1875	0.14	0.06	3.23	4.69	1.61	50000
12	NA482-472D	2.7500	4.7244	2.1250	2.5626	0.14	0.03	3.27	4.49	1.52	23100
13	NA484-472D	2.7559	4.7244	2.1250	2.5626	0.14	0.03	3.27	4.49	1.52	23100
14	NA567-563D	2.8750	5.0000	2.5625	3.1875	0.27	0.06	3.70	4.69	1.61	32000
15 *	NA495A-493D	3.0000	5.3750	2.1250	2.7500	0.14	0.03	3.62	5.12	1.31	25000
16	NA659-654D	3.0000	6.0000	3.0000	3.7500	0.14	0.06	3.66	5.55	1.43	39600
17	NA580-572D	3.2500	5.5115	2.6250	3.2500	0.14	0.03	3.86	5.24	1.45	51100
18	NA749-742D	3.2500	6.1250	3.3750	4.0000	0.14	0.06	3.90	5.63	1.80	50800

* INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).

2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
19 *	NA0593-592D	3.5000	6.0000	2.5000	3.2500	0.14	0.03	4.09	5.67	1.32	34800
20	NA759-752D	3.5000	6.3750	3.3750	4.1250	0.14	0.06	4.17	5.91	1.71	52300
21	NA52375-52637D	3.7500	6.3750	2.4375	3.2499	0.14	0.03	4.41	6.06	1.23	36100
22	NA776-773D	3.7500	7.0866	3.3750	4.1250	0.14	0.03	4.49	6.61	1.51	88200
23	NA776-774D	3.7500	7.1250	3.3750	4.1250	0.14	0.06	4.49	6.61	1.51	88200
24	NA691-672D	4.0000	6.6250	2.7500	3.6250	0.14	0.03	4.65	6.30	1.24	68200
25	NA780-773D	4.0000	7.0866	3.3750	4.1250	0.14	0.03	4.69	6.61	1.51	88200
26	NA780-774D	4.0000	7.1250	3.3750	4.1250	0.14	0.06	4.69	6.61	1.51	88200
27 *	NA861-854D	4.0000	7.5000	4.0000	5.0000	0.14	0.06	4.72	6.85	1.74	125400
28	HH221449NA-HH221410D	4.0000	7.5000	4.1250	5.0000	0.14	0.06	4.80	7.05	1.74	85200
29	NA782-773D	4.1250	7.0866	3.3750	4.1250	0.14	0.03	4.80	6.61	1.51	88200
30	NA782-774D	4.1250	7.1250	3.3750	4.1250	0.14	0.06	4.80	6.61	1.51	88200
31	NA71450-71751D	4.5000	7.5000	3.1875	4.1875	0.14	0.06	5.20	7.13	1.40	58300
32	NA938-932D	4.5000	8.3750	4.6250	5.6250	0.14	0.06	5.28	7.60	1.79	142900
33	NA48291-48220D	5.0000	7.1875	2.8750	3.3750	0.14	0.03	5.55	6.93	1.91	43200
34	NA798-792D	5.0000	8.1250	3.2500	4.2500	0.14	0.03	5.71	7.80	1.27	97800
35	NA95500-95927D	5.0000	9.2500	4.5000	5.6250	0.14	0.06	5.87	8.54	1.58	153000
36	NA48390-48320D	5.3750	7.5000	2.8750	3.3750	0.14	0.03	5.94	7.24	1.82	75000

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TABLE I. DASH NUMBERS AND DIMENSIONS –Continued

DASH NUMBER	PART NUMBER 1/ CONE–CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
37 *	NA81550–81963D	5.5000	9.6250	3.1250	4.2500	0.14	0.06	6.34	9.02	1.66	111200
38	NA48686–48620D	5.6250	7.8750	2.8750	3.6876	0.14	0.03	6.22	7.60	1.74	74600
39	NA82576–82951D	5.7500	9.5000	4.1875	5.1875	0.14	0.06	6.54	8.90	1.32	82500
40	NA99600–99102D	6.0000	10.0000	4.3750	5.6250	0.14	0.06	6.85	9.37	1.43	156800
41	NA67787–67720D	6.8750	9.7500	3.3125	4.0625	0.14	0.03	7.56	9.45	1.33	66800
42	NA87700–87112D	7.0000	11.1250	3.1250	4.2500	0.14	0.06	7.87	10.50	1.45	119700
43	NA94700–94114D	7.0000	11.3750	4.3750	5.6250	0.22	0.06	7.99	10.71	1.25	173800
44	HM237545NA–HM237510D	7.0000	11.3750	4.3750	5.6250	0.22	0.06	7.95	10.68	1.83	211900
45	NA94700–94118D	7.0000	11.7500	4.3750	5.6250	0.22	0.06	7.99	10.71	1.25	173800
46	H239649NA–H239612D	7.3750	12.6250	5.4375	7.3125	0.22	0.06	8.43	11.73	1.83	303000
47	NA93800–93127D	8.0000	12.5000	4.3750	5.7500	0.22	0.06	9.06	11.81	1.12	200400
48 *	NA130902–131401D	9.0000	14.0000	4.3750	5.7500	0.25	0.06	10.08	13.01	1.77	231800
49	NA170950–171451D	9.5000	14.5000	3.3750	4.7500	0.25	0.06	10.59	13.27	1.61	165300

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TABLE II

BEARINGS MANUFACTURED WITH CORRECT RUNNING CLEARANCE FOR THESE FITS										
DASH NUMBER (REF)	LIVE SHAFT DIAMETER LIMITS	DEAD SHAFT DIAMETER LIMITS	HOUSING		DASH NUMBER (REF)	LIVE SHAFT DIAMETER LIMITS	DEAD SHAFT DIAMETER LIMITS	HOUSING	
	TIGHT FIT	SPLIT FIT	LIVE SHAFT BORE LIMITS	DEAD SHAFT BORE LIMITS		TIGHT FIT	SPLIT FIT	LIVE SHAFT BORE LIMITS	DEAD SHAFT BORE LIMITS
			LOOSE FIT	TIGHT FIT				LOOSE FIT	TIGHT FIT
01	0.7515–0.7510	0.7505 – 0.7500	1.8534–1.8524	1.8494–1.8484	27	4.0030–4.0020	4.0010 — 4.0000	7.5030–7.5020	7.4990–7.4980
02	1.2515–1.2510	1.2505 – 1.2500	2.3155–2.3145	2.3115–2.3105	28	4.0030–4.0020	4.0010 — 4.0000	7.5030–7.5020	7.4990–7.4980
03	1.5015–1.5010	1.5005 – 1.5000	2.7200–2.7190	2.7160–2.7150	29	4.1280–4.1270	4.1260 — 4.1250	7.0896–7.0886	7.0856–7.0846
04	1.7517–1.7510	1.7505 – 1.7500	3.7530–3.7520	3.7490–3.7480	30	4.1280–4.1270	4.1260 — 4.1250	7.1280–7.1270	7.1240–7.1230
05	1.9700–1.9695	1.9690 – 1.9685	3.5463–3.5453	3.5423–3.5413	31	4.5035–4.5025	4.5010 — 4.5000	7.5030–7.5020	7.4990–7.4980
06	2.0020–2.0010	2.0005 – 2.0000	3.6748–3.6738	3.6708–3.6698	32	4.5035–4.5025	4.5010 — 4.5000	8.3780–8.3770	8.3740–8.3730
07	2.0020–2.0010	2.0005 – 2.0000	4.2530–4.2520	4.2490–4.2480	33	5.0035–5.0025	5.0010 — 5.0000	7.1905–7.1895	7.1865–7.1855
08	2.1270–2.1260	2.1255 – 2.1250	4.3780–4.3770	4.3740–4.3730	34	5.0035–5.0025	5.0010 — 5.0000	8.1280–8.1270	8.1240–8.1230
09	2.1674–2.1664	2.1659 – 2.1654	3.9400–3.9390	3.9360–3.9350	35	5.0035–5.0025	5.0010 — 5.0000	9.2530–9.2520	9.2490–9.2480
10	2.3770–2.3760	2.3755 – 2.3750	4.8780–4.8770	4.8740–4.8730	36	5.3785–5.3775	— —	7.5030–7.5020	—
11	2.6275–2.6265	2.6255 – 2.6250	5.0030–5.0020	4.9990–4.9980	37	5.5040–5.5030	— —	9.6280–9.6270	—
12	2.7525–2.7515	2.7505 – 2.7500	4.7274–4.7264	4.7234–4.7224	38	5.6290–5.6280	— —	7.8780–7.8770	—
13	2.7584–2.7574	2.7564 – 2.7559	4.7274–4.7264	4.7234–4.7224	39	5.7540–5.7530	— —	9.5030–9.5020	—
14	2.8775–2.8765	2.8755 – 2.8750	5.0030–5.0020	4.9990–4.9980	40	6.0040–6.0030	— —	10.0030–10.0020	—
15	3.0025–3.0015	3.0005 – 3.0000	5.3780–5.3770	5.3740–5.3730	41	6.8795–6.8785	— —	9.7530–9.7520	—
16	3.0025–3.0015	3.0005 – 3.0000	6.0030–6.0020	5.9990–5.9980	42	7.0045–7.0035	— —	11.1280–11.1270	—
17	3.2525–3.2515	3.2505 – 3.2500	5.5145–5.5135	5.5105–5.5095	43	7.0045–7.0035	— —	11.3780–11.3770	—
18	3.2525–3.2515	3.2505 – 3.2500	6.1280–6.1270	6.1240–6.1230	44	7.0045–7.0035	— —	11.3780–11.3770	—
19	3.5030–3.5020	3.5010 – 3.5000	6.0030–6.0020	5.9990–5.9980	45	7.0045–7.0035	— —	11.7530–11.7520	—
20	3.5030–3.5020	3.5010 – 3.5000	6.3780–6.3770	6.3740–6.3730	46	7.3795–7.3785	— —	12.6310–12.6290	—
21	3.7530–3.7520	3.7510 – 3.7500	6.3780–6.3770	6.3740–6.3730	47	8.0050–8.0040	— —	12.5060–12.5040	—
22	3.7530–3.7520	3.7510 – 3.7500	7.0896–7.0886	7.0856–7.0846	48	9.0055–9.0045	— —	14.0060–14.0040	—
23	3.7530–3.7520	3.7510 – 3.7500	7.1280–7.1270	7.1240–7.1230	49	9.5060–9.5050	— —	14.5060–14.5040	—
24	4.0030–4.0020	4.0010 – 4.0000	6.6280–6.6270	6.6240–6.6230	—	—	— —	—	—
25	4.0030–4.0020	4.0010 – 4.0000	7.0896–7.0886	7.0856–7.0846	—	—	— —	—	—
26	4.0030–4.0020	4.0010 – 4.0000	7.1280–7.1270	7.1240–7.1230	—	—	— —	—	—

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REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality – carburizing Grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295, ASTM - A534, or ASTM-A535. Cage: carbon steel (one piece stamped). Powdered process steel may be used.

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range B		Tolerance	
Over	Incl.	Plus	Minus
0 3.0000	3.0000 12.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 5.0000	5.0000 12.0000	100 300	0 0

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

4. Operating temperature: Recommended operating temperature not to exceed 230° F.

5. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-197.

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6. Contact angle: All bearings are normal angle bearings having a contact angle (α) between 10 and 19 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

7. Part number: The part number consist of the letters FFB, the basic federal specification number, the specification sheet number and a dash number from table I.

EXAMPLE: FFB187/08 - 25

- 25 — Dash Number
- FFB187/08 — Federal Specification Number

8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

9. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: Inch units can be converted to soft metric units as follows:

Inch	X	25.4	=	millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS	MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19088-1	FFB187/08-01	MS19088-26	FFB187/08-12
MS19088-2	FFB187/08-02	MS19088-27	FFB187/08-13
MS19088-3	FFB187/08-03	MS19088-28	FFB187/08-14
MS19088-4	FFB187/08-04	MS19088-29	FFB187/08-21
MS19088-5	FFB187/08-06	MS19088-30	FFB187/08-22
MS19088-6	FFB187/08-07	MS19088-31	FFB187/08-23
MS19088-7	FFB187/08-10	MS19088-32	FFB187/08-24
MS19088-8	FFB187/08-15	MS19088-33	FFB187/08-25
MS19088-9	FFB187/08-16	MS19088-34	FFB187/08-29
MS19088-10	FFB187/08-19	MS19088-35	FFB187/08-30
MS19088-11	FFB187/08-20	MS19088-36	FFB187/08-33
MS19088-12	FFB187/08-26	MS19088-37	FFB187/08-38
MS19088-13	FFB187/08-27	MS19088-38	FFB187/08-39
MS19088-14	FFB187/08-31	MS19088-39	FFB187/08-41
MS19088-15	FFB187/08-32	MS19088-40	FFB187/08-42
MS19088-16	FFB187/08-34	MS19088-41	FFB187/08-44
MS19088-17	FFB187/08-35	MS19088-42	FFB187/08-45
MS19088-18	FFB187/08-37	MS19088-43	FFB187/08-46
MS19088-19	FFB187/08-40	MS19088-44	FFB187/08-49
MS19088-20	FFB187/08-43	-	-
MS19088-21	FFB187/08-47	-	-
MS19088-22	FFB187/08-48	-	-
MS19088-23	FFB187/08-05	-	-
MS19088-24	FFB187/08-08	-	-
MS19088-25	FFB187/08-09		

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